

SECTION I – PROJECT NARRATIVE

PREDICTING OPEN SOURCE SOFTWARE DEVELOPERS BREAKS AND ITS IMPACTS

1 Rationale

Modern IT systems heavily rely on Open-Source Software (OSS), powering everything from critical infrastructure to everyday applications. These OSS initiatives are supported by developers who work voluntarily, shifting their focus to career changes, family obligations, or other priorities. In the same way, these projects are susceptible to issues like contributor burnout, resource scarcity, and a decline in engagement. As a result, numerous important open-source projects either stagnate with mounting maintenance backlogs or face complete abandonment.

Broader Impacts. This project addresses the challenge of contributors taking breaks from OSS projects by developing and evaluating a predictive modeling framework to predict the impact of developer breaks. Grounded in a dynamic sociotechnical framework, the predictive model will use contribution patterns, collaboration behaviors, and communication traces to provide maintainers with early warnings and actionable insights for sustaining project development.

Intellectual Merit. This project advances the understanding of contributor break in OSS projects by systematically investigating it as a sociotechnical phenomenon and by gaining knowledge in developing a tool. The research combines empirical studies with validated instruments to model the sociotechnical factors driving the breaks. The predictive framework will provide a transformative contribution to the field, equipping OSS maintainers with tools to address sustainability challenges proactively. Our proposal integrates predictive analytics, impact assessment, and real-world assessment, ensuring the results are both solid and practical. Additionally, the project addresses gaps in existing research by focusing on long-term contributors and maintainers, whose breaks poses significant risks to OSS projects.

2 Expertise

The proposed project is well-suited to our team’s combined expertise and collaborative history. Our team combines expertise in open source software research, statistical modeling, and machine learning, ensuring a robust approach to addressing the challenges of OSS contributor breaks.

PI Steinmacher is a leading expert in onboarding, retention, and breaks in OSS. His research spans critical areas, including newcomer integration [28, 29, 34, 27, 31, 22, 33], break [7, 15], and the role of maintainers in OSS [11]. He has also explored the educational

applications of OSS [21, 20, 23] and has extensive experience mining software repositories, applying statistical modeling and machine learning techniques in the OSS domain [40, 41, 36, 37, 24, 25], and conducting CSCW research [35, 5, 30, 39].

The PhD student who will work on this project, **Pedro Oliveira**, brings additional expertise in data processing and analysis for OSS. His technical proficiency will contribute to data collection, curation, and building and evaluating the prediction model. His studies include analyzing OSS activity and maintenance signals before, during, and after OSS death [26], and using social network analysis to understand community interaction during OSS death [3].

3 Research Questions

Open Source Software (OSS) forms the backbone of our digital society [12]. It has made technology more accessible globally, empowering companies and developers. However, its decentralized, peer-production nature complicates contributor retention and the securing of financial support for developers [42, 13]. Many organizations use OSS without contributing to its maintenance, jeopardizing its sustainability. Despite these challenges, OSS remains a cornerstone of the modern economy [14]. For example, in 2016, the internet was momentarily disrupted because 11 lines in the OSS JavaScript registry npm were unpublished [16]. This incident is evidence of the reliance of major organizations such as Facebook, Netflix, Airbnb, and the US government [8, 38] on OSS.

A key component of OSS sustainability is fostering a strong and enduring community, which depends on attracting and retaining contributors. Despite significant research into onboarding newcomers and reducing barriers to entry [33, 30, 32], the dynamics of contributor breaks, particularly among core contributors or maintainers, remain under-explored. The challenges of tracking and managing breaks are amplified in OSS settings compared to traditional industry environments, due to loosely coupled work patterns, reduced awareness among co-workers, and less centralized hierarchical control and organization [9].

Understanding developers' breaks is essential to maintaining OSS projects' sustainability [2, 18]. Without effective tools to anticipate and mitigate these risks, maintainers often struggle to manage disruptions caused by unexpected contributor breaks [15]. Addressing these gaps is essential for ensuring the sustainability of OSS projects and fostering resilient development communities.

The problem is especially acute when maintainers or core developers take extended breaks. When this happens, projects suffer from knowledge silos, stalled progress, and disruptions to workflows, such as code reviews, contributions, and mentorship. High attrition rates among core contributors can lead to stalled project evolution and increase the maintenance burden on remaining developers [4, 2].

Break, in our context, refers to both short breaks that developers take from the project (coming back later) and when developers leave the project for good. Breaks are particularly concerning in critical infrastructure projects, where losing expertise can leave security vulnerabilities unresolved and create risks for dependent systems [7]. The abrupt departure of a maintainer responsible for a widely used module can delay updates, reduce

user confidence, and harm the ecosystem’s long-term viability. Proactively addressing these challenges requires strategies to understand and assess the breaks, ensuring that projects remain sustainable over time.

This proposal addresses these issues through research questions centered around a prediction model designed to assist OSS maintainers with predictive insights and proactive management tools:

- How to create predictive models trained on contributor activity patterns to predict when developers are at risk of taking breaks?
- How can models assess the impact of a developer’s breaks on a project?

4 Research Approach

To effectively anticipate and mitigate developer breaks in open-source software (OSS) projects, we propose a structured approach that integrates data collection, predictive modeling, and validation. As illustrated in Figure 1, our methodology follows a multi-stage framework designed to (i) systematically capture developer activities, (ii) build predictive models for identifying early signs of breaks, and (iii) validate the predictive framework through statistical analysis and machine learning techniques. By leveraging a combination of empirical data, machine learning models, and theoretical validation, our approach aims to provide a comprehensive understanding of the factors influencing developer retention and break. The following sections outline the key stages of our methodology, detailing the steps taken to construct a robust break prediction framework.

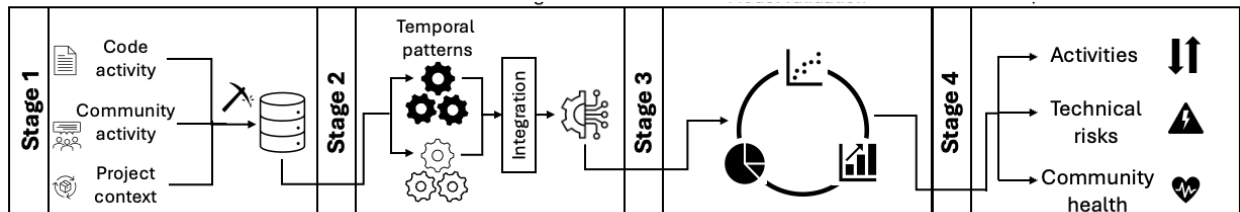


Figure 1: Model development and validation overview

Stage 1 establishes the foundation for our break prediction framework by creating a comprehensive dataset of developer activities. This stage encompasses three key steps: (i) establishing a project sample, (ii) collecting a rich set of activity traces, and (iii) contextualizing these activities within project-specific factors.

(i) To ensure the broad applicability of our approach, we will build a comprehensive dataset of OSS projects. For this proposal, we will adopt a stratified random sampling approach to capture diverse contribution patterns beyond GitHub-centric projects. Our sampling framework will consider three key dimensions: core project characteristics (including domain, size, age, and programming languages), community aspects (such as governance model and corporate involvement), and development infrastructure (spanning platforms and communication channels). We will include abandoned projects in our

sample, as these could provide valuable historical insights into break patterns and factors that may have led to project discontinuation.

(ii) We will collect data that enables us to go beyond the basic metrics, as individual contribution rhythms through coding activities (commits, pull requests) and non-coding activities (issues and pull request discussions) in GitHub. We will introduce new categories of metrics that can serve as early warning indicators of potential breaks. This expanded set of metrics will provide a more comprehensive view of developer engagement patterns and potential early breaks signals [17, 18].

(iii) Finally, we will contextualize these individual activity traces within broader project ecosystem factors that may influence break patterns. This includes analyzing project dependency networks [10] to understand maintenance complexity, examining project maturity indicators such as size and age [19], and considering organizational factors like corporate sponsorship [42, 6].

Stage 2 focuses on developing a predictive model to anticipate TF developer breaks using the activity traces collected in the previous stage. Our proposed approach combines multiple analysis techniques to create a robust prediction framework that addresses the potential limitations of other previous far-out values approach [7].

To build the enhanced break prediction model, we will implement two strategies: building customized models for individual projects and developing a generalized cross-project solution. To build the customized models, we will first establish project-specific baseline activity patterns for each TF developer through a combination of complementary methods. For the generalized solution, we will develop a general pre-trained model that can work across projects, enabling quick risk assessment even for projects with limited historical data or those wanting to avoid data collection and model training overhead.

Stage 3 involves two complementary phases of statistical analysis to validate and refine our prediction model’s technical performance and theoretical soundness. Following standard machine learning practices, we will partition the collected data into training, validation, and test sets to ensure robust evaluation of our model’s predictions.

We will evaluate our model by comparing its predicted risk levels and timing of breaks against actual developer break patterns, measuring how accurately it anticipates both the likelihood and the time window.

Additionally, we will validate that our model performs consistently across different developer groups and project contexts. Implementing a verification step can enhance the accuracy and reliability of the model, leading to better results.

We will develop regression models to examine how different factors contribute to break likelihood. These analyses will incorporate directly measurable variables (such as commit frequency and issue resolution times) and composite activity metrics, which we will construct through dimensionality reduction techniques (such as Principal Component Analysis [1]).

These techniques will help us identify meaningful patterns in developer behavior (e.g., ‘review engagement’ combining review frequency, length, and response time, or ‘code contribution intensity’ incorporating commit size, frequency, and complexity) while reducing dimensionality and avoiding multicollinearity.

5 Follow-on plan

The primary deliverable is a comprehensive break prediction framework combining time series modeling and impact assessment capabilities. To help OSS projects analyze contributor activity patterns and predict breaks, all the source code employed to build the model and the model parameters will be released as open source.

All the analytical components will be modular, allowing communities to adapt and extend them based on their specific needs and data sources. The general pre-trained model will be uploaded to Hugging Face or a similar platform for easy access. We will release an open dataset enabling researchers to conduct further analyses and studies in related domains.

Since this is a new potential direction for work with open source projects, having a good track record of studies related to breaks and turnover is important to creating a reputation on the topic. The goal is that partial results would be used as preliminary work to seek external funding, specifically focusing on NSF CISE Core solicitation (\$600,000 total, with around \$400,000 of direct costs) and open source and open science-related calls (Sloan Foundation Technology projects; NSF FAIROS and CSSI).

6 Funding track record

Dr. Steinmacher does not have any track record of intramural grants. However, the work that was when Dr. Steinmacher had his startup funds converted into an NSF Core Large grant (2018–2021), and another spin-off resulted in an NSF HSI:IUSE grant (2023–2026). Both of these grants are related to Open Source, focusing on (i) onboarding and inclusion in projects; and (ii) student OSS learning experience. This grant may result in preliminary studies to inform future submissions related to the sustainability of software projects and early identification of issues caused by developers' turnover.

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Section II – ADDITIONAL DOCUMENTATION